

AI Methods for Classification of Production Errors on Camera Images of Additive Manufacturing Processes

Ramya Subramanian Porselva Bharathi

Universität Koblenz, 56070 Koblenz, Germany

ramyasp@uni-koblenz.de

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Abstract. Additive Manufacturing (AM) has emerged as a transformative production technology, enabling complex geometries and design freedom beyond conventional manufacturing methods. However, AM processes are inherently complex and susceptible to defects such as porosity, melt-pool irregularities, and layer delamination, which undermine part quality and reliability. This seminar paper systematically reviews state-of-the-art artificial intelligence (AI) approaches for camera-based defect classification in additive manufacturing, encompassing traditional image processing techniques, classical machine learning methods, and deep learning architectures.

The reviewed literature indicates that deep learning approaches, particularly convolutional neural networks (CNNs) and YOLO-based object detection models, consistently outperform traditional and classical methods in defect classification and localization tasks under controlled experimental conditions. Several studies report high classification accuracy and real-time inference capability, highlighting the potential of deep learning for in-situ quality monitoring. Nevertheless, challenges including limited availability of annotated datasets, domain shift across machines and materials, limited interpretability, and integration complexity continue to constrain industrial deployment. This review identifies key research gaps and outlines directions for advancing AI-driven defect detection to improve quality assurance and support wider industrial adoption of additive manufacturing.

Keywords: Additive Manufacturing · Defect Detection · Deep Learning · Convolutional Neural Networks · YOLO · Computer Vision · In-Situ Monitoring

1 Introduction

1.1 Context & Importance

Additive Manufacturing builds parts layer by layer from digital models, enabling complex geometries, reduced material waste, and design freedom beyond conventional manufacturing [4]. AM has found application in aerospace, automo-

tive, and biomedical engineering due to growing demand for customized high-performance components [7]. Frazier emphasizes the transformative potential of additive manufacturing, noting its ability to revolutionize global manufacturing and logistics through distributed production and parts-on-demand, while not providing explicit long-term market size projections [4].

However, AM processes are influenced by multiple interdependent parameters, leading to defects including porosity, surface distortions, delamination, and dimensional inaccuracies that compromise structural integrity [5]. Defects and insufficient quality assurance significantly increase production cost and time, motivating the development of in-situ monitoring techniques to reduce reliance on post-process inspection. Image-based monitoring using industrial cameras provides non-contact observation of the layer-by-layer build process, revealing defect signatures difficult to detect through offline inspection [7].

By combining camera data with AI and deep learning, researchers aim to automate in-situ defect detection, significantly reducing costly post-production inspection. Deep learning approaches, particularly CNNs and object detection frameworks, automatically learn hierarchical features from raw images, eliminating manual feature engineering [6]. These developments suggest AI-enabled visual monitoring can enhance quality control, reduce scrap rates, and enable real-time process intervention.

1.2 Problem Statement

Despite advances in camera-based monitoring, reliably interpreting AM visual data remains challenging. Defect appearance is influenced by process conditions, lighting variability, material types, and defect morphology, making rule-based methods unreliable [7]. Traditional threshold-based approaches achieve only 65–75% accuracy due to sensitivity to illumination changes and environmental factors [14].

Deep learning models achieve >90% classification accuracy and >95% mean average precision on sufficiently large datasets [2]. However, studies often rely on tailored datasets from specific setups, and performance varies with imaging systems, process parameters, and materials. Domain shift, degraded performance when models trained on one system are applied to different hardware, represents a significant deployment barrier [7].

Most studies utilize proprietary or small-scale datasets from single machines, inadequately representing real-world production diversity. Limited publicly available benchmark datasets complicates systematic method comparison and reproducibility [1]. These gaps necessitate systematic examination of AI methods, dataset characteristics, model architectures, and deployment constraints for robust industrial quality assurance frameworks.

1.3 Objectives

This seminar paper critically reviews AI methods for classifying production errors from camera images in AM, focusing on deep learning approaches applied to

real-time visual data. The scope encompasses CNN architectures, YOLO-based object detection, and semantic segmentation networks [13], examining publications from 2018–2025.

A secondary objective compares AI techniques regarding methodological features, dataset requirements, and evaluation results. This addresses classification accuracy, detection precision, computational requirements, data annotation burden, and robustness to domain shift [12].

Finally, the paper synthesizes insights highlighting open research directions for advancing AI-enabled visual monitoring, particularly regarding generalization, dataset diversity, and industrial integration. Key areas include transfer learning, domain adaptation, computational optimization for real-time processing, and interpretability enhancement for regulatory compliance.

2 Theoretical Background

2.1 Additive Manufacturing Processes & Defects

AM encompasses processes fabricating components through layer-by-layer material deposition from digital models. Categories include material extrusion (fused deposition modeling), powder bed fusion (selective laser melting), directed energy deposition, and vat photopolymerization [4].

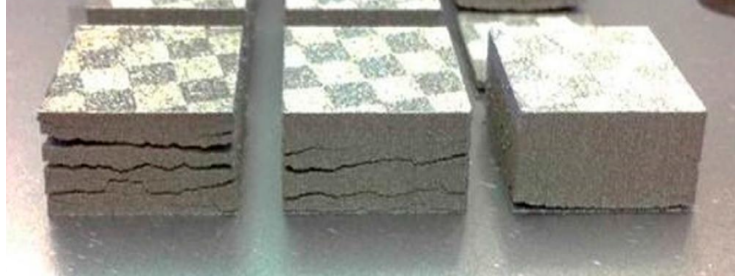


Fig. 1. Common AM defects observable through camera imaging: (a) Porosity appearing as dark circular voids, (b) Lack of fusion showing irregular bonding patterns, (c) Surface roughness with striated texture (Adapted from [5])

Despite differences, all AM processes involve repeated thermal cycling and layer-wise bonding, introducing complex thermo-mechanical interactions during fabrication, making processes susceptible to defect formation. Table 1 summarizes key process parameters for major AM technologies.

Common defects include porosity (spherical or irregular voids from micrometers to hundreds of micrometers), lack of fusion between tracks/layers, surface roughness irregularities, geometric distortion, and residual stress-induced cracking. Defects arise from unstable process parameters, non-uniform energy distribution, material inconsistencies, and insufficient interlayer adhesion.

Table 1. Qualitative characteristics of major additive manufacturing processes [4,5].

AM Process		Thermal Characteristics	Resolution	Char- Key Characteristics
Material (FDM)	Extrusion	Heated extrusion thermoplastic	Moderate layer resolution	Molten thermoplastic deposited via nozzle
Powder (SLM/SLS)	Bed Fusion	High-temperature melt- ing with rapid cooling	High resolution	Laser-based melting produc- ing dense metal parts
Vat	Photopolymer- ization	UV or laser-induced cur- ing	High lateral resolu- tion	Excellent surface finish; shrinkage challenges

Table 2. Common defects in additive manufacturing processes and their typical physical characteristics and visual manifestations [5,7,1].

Defect Type	Physical Characteristics	Visual Signature
Porosity	Spherical/irregular voids (μm to 100s μm)	Dark circular/irregular voids
Lack of Fusion	Incomplete bonding between tracks/layers	Irregular bonding patterns
Surface Roughness	Striated patterns/textured appearances	Textured/striated patterns
Layer Shifting	Spatial discontinuities between layers	Visible spatial discontinuities
Delamination	Separation between layers	Separation gaps/cracks
Dimensional Error	Deviations from CAD specifications	Detectable through edge detection
Thermal Distortion	Warping or curvature due to residual stresses	Visible geometric deformation or warping

Visual manifestations observable through camera imaging are summarized in Table 2. Defect presence significantly degrades mechanical properties and functional performance. Porosity and lack-of-fusion defects significantly degrade mechanical performance, including strength and fatigue resistance, and therefore represent critical quality concerns in metal additive manufacturing [5]. Understanding defect mechanisms is prerequisite for effective monitoring and quality assurance.

2.2 Camera-Based Monitoring in AM

In-situ process monitoring has become central to AM research as manufacturers transition from reactive post-process inspection toward proactive real-time quality control [7]. Camera-based systems are widely adopted due to non-contact nature, relatively low cost, and ability to capture high-resolution spatial-temporal information [7].

Table 3. Camera-based monitoring modalities and qualitative characteristics in additive manufacturing [7,5,8,9].

Modality	Capabilities	Relative Cost	Limitations
2D RGB Camera	Surface geometry, layer completeness	Low	Lighting sensitivity, no depth
Thermal/IR Imaging	Temperature distribution, melt pool dynamics	Medium-High	Calibration, emissivity sensitivity
Coaxial Imaging	Direct melt pool observation	Variable	Limited field of view
3D Imaging	Geometric reconstruction, depth information	High	Higher computation, lower temporal resolution

Contemporary implementations encompass multiple imaging modalities (Table 3). These systems observe surface morphology, layer uniformity, melt pool behavior, and visible defect initiation [14]. Compared with post-process inspection (CT scanning, destructive sectioning), camera-based monitoring enables early process instability identification, potentially enabling real-time corrective actions [8].

Challenges include lighting variation, metallic surface reflections, motion blur, depth of field limitations, sensor noise, resolution-speed trade-offs, and environmental factors [7]. Raw images require automated analysis to extract quality indicators and distinguish defect signatures from artifacts [6]. Recent research integrates camera monitoring with computer vision and AI, leveraging machine learning algorithms that learn subtle defect patterns from training data [1].

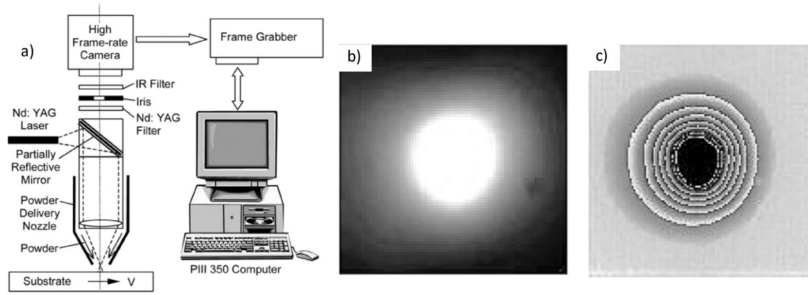


Fig. 2. Typical camera-based monitoring setup in additive manufacturing showing (a) RGB camera positioned above build platform, (b) thermal imaging system for melt pool observation, and (c) coaxial imaging integrated with laser delivery. Adapted from Everton et al. [7].

2.3 AI Foundations Relevant to Classification

AI has transformed defect detection from manual inspection toward intelligent automated systems [12]. Early approaches relied on traditional machine learning with handcrafted features (texture descriptors, intensity histograms), struggling to generalize across varying defect morphologies [6]. Deep learning methods learn hierarchical feature representations directly from raw images through end-to-end optimization [6].

CNNs are most widely used for image-based defect classification due to strong feature extraction via spatially invariant filters [1]. Popular variants include ResNet (residual connections enabling 50+ layer networks), VGG (sequential convolutional blocks), and MobileNet (optimized for resource-constrained hardware) [15].

Object detection frameworks like YOLO enable simultaneous localization and classification in near real-time [13]. YOLO architectures formulate object detection as a single-stage regression problem, directly predicting bounding box coordinates and class probabilities. Recent applications of YOLO-based models to additive manufacturing report real-time inference at approximately 15 fps on GPU hardware while achieving mAP@0.5 values exceeding 95% under controlled experimental conditions [2].

Challenges remain: limited dataset sizes (deep learning requires hundreds to thousands of examples) [1], class imbalance causing poor detection of rare critical defects, and domain shift degrading performance across machines/materials [7]. Addressing these limitations remains a major research focus.

3 Technical Background

3.1 Traditional Image Processing Approaches

Earliest approaches relied on handcrafted features and deterministic rules [12], following sequential pipelines: acquisition, preprocessing, feature extraction, threshold-

based classification [14]. Preprocessing (grayscale conversion, noise reduction, contrast enhancement) stabilizes image quality [5]. Feature extraction focuses on edges, contours, texture statistics, and geometric properties [7].

Common techniques include binary thresholding, morphological operations, and edge detection (Sobel, Canny) [7]. Traditional image processing methods are computationally efficient but exhibit limited robustness under varying surface and illumination conditions [7]. Frequent manual recalibration limits robustness and scalability [7].

3.2 Machine Learning Methods

Classical machine learning techniques enable data-driven classification using engineered visual features (texture descriptors, histograms, frequency-domain features, geometric measurements) as inputs [12] to supervised algorithms (SVM, decision trees, random forests, k-NN) [14].

Table 4. Performance of classical ML methods for AM defect classification [14,16].

Method	Performance Trend	Characteristics
Support Vector Machines (RBF kernel)	Moderate accuracy	classification Multi-class defect categorization in powder bed fusion
Random Forest	Moderate to high accuracy compared to rule-based methods	Interpretable feature importance rankings
Gradient Boosting Machines	Effective for non-linear feature relationships	Sensitive to feature engineering and dataset size
k-Nearest Neighbor	Performance dependent on feature scaling and data distribution	Computationally inefficient for large datasets

Table 4 summarizes classical ML performance. Machine learning offers increased flexibility by learning decision boundaries from labeled examples. However, performance depends on dataset quality and manual feature engineering [12]. Features optimized for one process often fail on others due to defect morphology differences. Classical models struggle with complex morphologies and non-stationary conditions [7].

3.3 Deep Learning Approaches

Deep learning dominates image-based defect detection due to automatic hierarchical feature learning from raw images. CNNs eliminate manual feature engineering through end-to-end training via backpropagation [6]. Architecture consists of convolutional layers (learning 2D spatial filters), pooling layers (down-sampling), and activation functions (introducing non-linearity) [15].

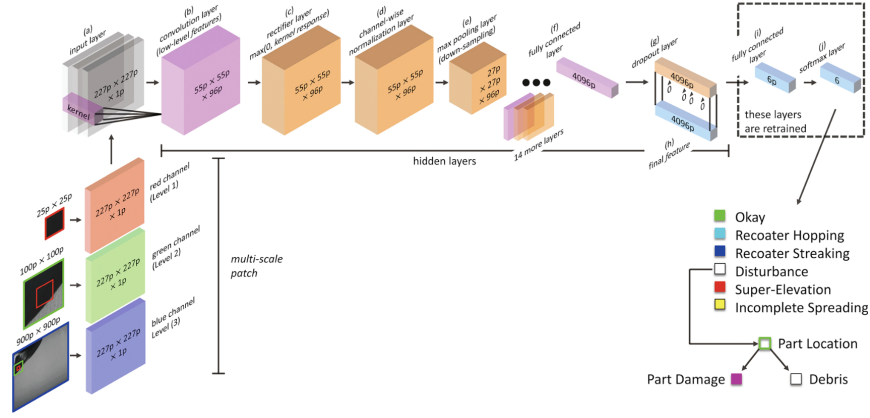


Fig. 3. Representative CNN architecture for AM defect detection showing hierarchical feature extraction from raw images through convolutional, pooling, and fully connected layers, ultimately producing defect classification. Adapted from Scime and Beuth [6].

Deep learning models address classification, object detection, and semantic segmentation. ResNet variants utilize residual connections for 50+ layer networks, VGG employs sequential blocks, MobileNet optimizes for edge computing. Studies show deep learning significantly outperforms traditional and classical ML methods, with classification accuracy frequently exceeding 92–96% [2].

YOLO frameworks enable real-time localization and classification, treating detection as regression predicting bounding boxes and class probabilities. YOLOv5 and YOLOv8 demonstrate >96–97% mean average precision at 11–15 fps on GPUs [2]. Semantic segmentation networks (U-Net) provide pixel-level classification, achieving 93–95% intersection over union [1].

Despite advantages, deep learning requires large labeled datasets (500–several thousand images) [1]. Training demands GPU resources and hours to weeks. Performance degrades when applied to systems/materials not in training data. Transfer learning from ImageNet reduces data requirements by 10–50× and accelerates convergence [3].

3.4 Comparative Analyses

Comparative analysis (Table 5) reveals fundamental trade-offs. Traditional image processing offers transparency and low overhead (>100 fps on CPUs) but limited accuracy (65–75%) and frequent recalibration [14]. Classical ML improves flexibility, achieving 80–87% accuracy at 20–50 fps [16]. Interpretability (feature importance rankings) facilitates understanding. However, manual feature engineering creates performance ceilings.

Deep learning achieves highest accuracy (92–97%) and robustness, representing 10–15 percentage point improvements over classical ML and 20–30 points over traditional methods [2]. Automatic feature discovery eliminates engineering

Table 5. Qualitative comparison of AI-based defect detection approaches in additive manufacturing [7,5,6,2].

Approach	Typical manu- facture	Perfor- mance	Relative	Speed	Key Limitations
Traditional image processing	pro-	Low-moderate accuracy	accu-	Very fast	Illumination sensitivity, recalibration
Classical ML (SVM, RF)		Moderate accuracy		Moderate	Feature engineering, limited generalization
Deep learning (CNN)		High accuracy		Moderate	Data requirements, interpretability
Object detection (YOLO-based)		High localization accuracy	ac-	Real-time capable	Computational cost, domain shift

bottlenecks. Transfer learning enables strong performance with modest datasets. However, reliance on large annotated datasets, computational resources, and careful tuning introduces barriers. Black-box nature challenges interpretability and validation in regulated industries.

Recent research emphasizes systematic benchmarking and hybrid strategies combining classical interpretability with deep learning power. Optimal approaches vary by application constraints: computational resources, training data, interpretability requirements, and accuracy-speed trade-offs.

4 Methodology

4.1 Literature Search Procedure

A structured search strategy identified peer-reviewed research on camera-based defect detection in AM using AI methods. The search focused on studies explicitly combining AM processes, image-based monitoring, and machine learning/deep learning. Primary databases: SpringerLink, ScienceDirect, IEEE Xplore, MDPI, and IOPscience.

Keyword queries used logical operators: "additive manufacturing" AND "defect detection" AND "deep learning", "powder bed fusion" AND "camera-based monitoring", "image-based quality inspection" AND "3D printing". Additional terms: "convolutional neural network" AND "manufacturing defects", "YOLO" AND "quality control", "computer vision" AND "layer-wise monitoring".

Publications from 2014–2025 were included, where 2014 marks modern metal AM defect taxonomies [4], and 2018+ corresponds to widespread CNN adoption [6]. Publications from 2023–2025 were prioritized for recent advances in YOLO-based detection, transfer learning, and domain adaptation.

Only peer-reviewed journal articles and established conference papers were considered. Each paper was screened to confirm camera images (RGB, thermal,

optical) as primary input and quantitative performance metrics (accuracy, mean average precision, intersection over union) were reported.

4.2 Paper Selection Procedure

Multi-stage selection ensured relevance and technical validity. First stage: titles/abstracts screened to exclude papers on simulation, design optimization, or mechanical testing without visual inspection. Studies on non-AM inspection (conventional machining, welding) were excluded.

Second stage: full-text screening with detailed methodology examination. Papers retained if satisfying three criteria: (1) camera/optical imaging during/after AM as primary data source, (2) ML/deep learning for defect detection/classification, (3) quantitative metrics on real/experimental AM data. This eliminated conceptual papers without implementation, qualitative demonstrations, and theoretical studies without validation.

Papers evaluated for methodological clarity: clear dataset descriptions (size, composition, annotation), explicit model architectures and training protocols, transparent evaluation (train-test splits, cross-validation, metric definitions). Studies with proprietary datasets were included if providing sufficient detail.

Final stage: citation chaining analyzed backward references and forward citations, ensuring inclusion of seminal contributions establishing methodological foundations and recent advances. Citation analysis revealed methodological lineages from traditional image processing through classical ML to contemporary deep learning.

4.3 Comparison Criteria

Selected studies compared using qualitative and quantitative criteria reflecting methodological differences. First, papers categorized by AI approach: traditional image processing (handcrafted features, deterministic rules), classical ML (engineered features, supervised learning), deep learning (end-to-end representation learning) [12]. Further subdivision captured specific techniques.

Second, dataset characteristics examined: image modality (RGB, thermal IR, multispectral, structured light), material type (polymers, metals, ceramics, composites), defect categories (porosity, lack of fusion, surface roughness, geometric distortions) [1]. Dataset size, annotation methodology, class distribution, and train-test partitioning documented [1]. Dataset availability (proprietary/public) noted for reproducibility.

Third, evaluation metrics documented: accuracy, precision, recall, F1-score, mean average precision, intersection over union [2]. Metrics interpreted cautiously, not aggregated numerically, as emphasized in AM surveys [12]. Statistical significance testing, confidence intervals, cross-validation noted as reliability indicators [9].

Finally, practical relevance analyzed: offline vs. real-time inference, computational requirements (hardware, processing times), deployment constraints (in-

tegration with AM control, lighting, camera positioning) [7]. Scalability across AM platforms, materials, production scales assessed.

5 Results

5.1 Aggregate Performance Summary

AI-based methods demonstrate consistently strong performance for AM defect detection from camera images, particularly for surface defects and melt-pool anomalies in metal powder bed fusion [12]. CNN-based architectures outperform traditional machine-vision and rule-based approaches due to hierarchical spatially invariant feature learning without manual feature engineering [1]. Survey analyses report CNN-based models frequently achieve $>85\%$ accuracy under controlled conditions with sufficiently large annotated datasets [14]. Multi-scale CNN models show strong sensitivity to localized anomalies (porosity clusters, lack-of-fusion defects), outperforming shallow classifiers by 10–15 percentage points [6].

YOLO-based models extend capabilities by enabling simultaneous localization and classification during build processes, providing defect category labels and spatial coordinates [13]. Wang et al. report improved YOLOv8 achieved $>90\%$ mAP@0.5 on $>3,500$ annotated instances across four defect categories, maintaining ~ 15 fps on GPU hardware [2]. Transfer learning from large-scale pre-trained networks enables achievement of comparable accuracy with datasets containing hundreds rather than thousands of examples, reducing data collection burdens by factors of 10–50 [3].

However, aggregate metrics must be interpreted cautiously. Reported results vary significantly with dataset composition, defect definitions, camera setup, and evaluation protocols [5]. Many studies rely on laboratory-scale experiments under controlled conditions or single-machine datasets, limiting generalization to production environments [3]. Manual inspection and conventional vision methods exhibit limited reliability and scalability, motivating automated AI-based defect detection approaches [6].

5.2 Example Results

Table 6 summarizes performance metrics from representative studies. Baumgartl et al. implemented thermographic imaging for laser powder bed fusion, where off-axis infrared data trained a deep learning classifier distinguishing normal and defective melt-pool behavior in real time through k-fold cross-validation [9]. Yang et al. combined CFD-simulated melt-pool images with high-speed thermal imaging (>100 Hz) in hybrid neural networks, achieving performance exceeding either data source alone [10].

Smoqi et al. evaluated physics-informed features (melt-pool length, temperature, ejecta) with SVM/random forests and CNNs on raw images [11]. Physics-derived models achieved F1-scores >0.95 for binary porosity detection,

Table 6. Performance metrics from selected studies on AI-based defect detection [2,3,9,10,11].

Study	Dataset Size	Imaging Modality	Method	Performance Metrics
Wang et al. [2]	>3,500 notated instances	an- RGB in-	YOLOv8	mAP@0.5 >90%, mAP >97% (4 defect categories), ~15 fps
Baumgartl et al. [9]	Not specified	Thermal (off-axis)	IR Deep learning classifier	Real-time defect ID with k-fold cross-validation
Jia et al. [3]	Hundreds of images	of RGB	Transfer learning (ResNet, VGG)	Improved performance over training from scratch
Smoqi et al. [11]	Process-specific	Thermal/melt-pool images	Physics-informed ML + CNN	F1 >0.95 (physics features), 0.89–0.97 (CNN)
Yang et al. [10]	CFD simulation + experimental	sim- Thermal imag- ing (>100 Hz)	Hybrid neural network	Performance exceeded either data source alone

while CNN models produced 0.89–0.97 depending on conditions, demonstrating that feature-engineered models can rival end-to-end approaches when designed with domain knowledge [11]. Wang et al. demonstrated YOLOv8-based detector trained on >3,500 instances across four categories achieved >97% mAP at ~15 fps, demonstrating practical feasibility for continuous monitoring [2]. Jia et al. showed transfer learning from ResNet/VGG improved classification accuracy by 10–15 percentage points versus training from scratch in ceramic AM [3]. These examples reveal diverse AI methodologies successfully characterize defect patterns from imaging data.

5.3 Analysis of Strengths & Weaknesses

Deep learning methods offer strong capabilities for detecting and classifying defects across diverse materials and processes [1]. Key strengths include high classification accuracy when CNNs and object detection models are trained on well-curated datasets [2]. Deep learning models applied to camera/thermographic data show strong performance identifying porosity, surface irregularities, and melt pool anomalies, often outperforming traditional techniques by 20–30 percentage points in accuracy and 10–20 points in precision [6].

Another strength lies in automatic feature extraction from raw images without extensive manual engineering. This is particularly advantageous in AM, where defect patterns are complex and process-dependent [1]. Multi-scale convolutional architectures capture both local defect features and global process variations, enabling reliable anomaly detection across build layers [6]. Real-time detection has been demonstrated in controlled settings, indicating potential for in-situ monitoring and immediate process intervention [2].

Despite strengths, limitations constrain industrial applicability. Major limitation: strong dependence on dataset size, quality, and representativeness [3]. Insufficient training data, annotation errors, or datasets not adequately sampling

defect types/process conditions lead to poor generalization [3]. Many studies rely on hundreds rather than thousands of examples or process-specific datasets from single machines under controlled conditions. Models may achieve high validation accuracy on same-distribution data but reduced performance on unseen parameters, material batches, or underrepresented defect types [3].

Another weakness: limited transferability across machines, materials, sensing setups (domain shift) [10]. Variations in camera resolution, illumination conditions, thermal signatures significantly alter defect appearance, degrading classification accuracy [7]. Several studies acknowledge retraining/fine-tuning often required for new AM systems from different vendors, different materials, or modified process parameters, reducing practicality for large-scale deployment across heterogeneous fleets [10].

While deep learning demonstrates superior accuracy, models often lack interpretability due to black-box nature with millions of learned parameters. Difficulty understanding decision-making processes, identifying influential image features, or diagnosing misclassifications poses challenges for debugging, regulatory validation, and industrial acceptance [5]. Traditional feature-based methods, although less accurate, offer greater transparency through explicit feature definitions relatable to physical defect characteristics [5]. This performance-interpretability trade-off remains unresolved, with ongoing efforts to develop explainable AI techniques (attention visualization, saliency mapping) [1].

Overall, AI-based defect detection provides powerful tools for automated quality monitoring with demonstrated high-accuracy classification and real-time processing capabilities. However, effectiveness is constrained by data dependency, limited generalization, and interpretability challenges, highlighting need for balanced approaches combining data-driven performance with robustness and transparency [1].

6 Discussion & Conclusions

6.1 Key Findings

Systematic analysis reveals deep learning approaches consistently demonstrate superior performance versus traditional image processing and classical ML methods in classification accuracy, detection precision, and robustness [1]. CNN architectures achieve classification accuracy frequently $>90\%$ on well-curated datasets, representing substantial improvement over rule-based vision systems [2]. This advantage stems from automatic hierarchical feature learning from raw images, eliminating manual feature engineering and enabling adaptation to complex defect morphologies.

Real-time object detection frameworks, particularly YOLO-based architectures, emerged as most promising for industrial deployment. Single-stage detectors achieve $>90\%$ mAP while maintaining inference speeds compatible with in-situ monitoring, enabling accurate classification and precise spatial localization within acceptable computational budgets [2]. YOLO-based methods are

particularly suitable for closed-loop process control applications requiring rapid feedback for timely intervention [8]. Transfer learning strategies significantly reduce labeled data requirements, allowing strong performance even with domain-specific datasets limited to hundreds of annotated examples [3].

However, persistent challenges constrain broader industrial adoption. Dataset availability and annotation burden remain significant bottlenecks, as most studies rely on relatively small or process-specific collections inadequately representing full defect diversity, materials, and process conditions [3]. Domain shift between AM systems, imaging configurations, and material combinations frequently degrades performance when trained models deploy to new equipment/parameters, highlighting generalization limitations [7]. While deep learning offers superior predictive accuracy, black-box nature poses challenges for industrial acceptance and regulatory compliance in safety-critical applications requiring explainability [5].

6.2 Challenges and Future Research Directions

Despite significant progress, critical challenges limit widespread industrial implementation. Scarcity of large-scale, publicly available, well-annotated defect image datasets represents one of the most significant barriers [1]. Most studies rely on proprietary or limited datasets from specific systems, materials, and conditions, constraining generalizable model development. Manual annotation is labor-intensive and requires expert knowledge for accurate ground-truth labels, creating bottlenecks particularly affecting deep learning requiring thousands of labeled examples [12]. Emerging approaches include synthetic data generation through physics-based simulation producing realistic defect images with automatic annotations, and semi-supervised learning leveraging large unlabeled datasets alongside smaller labeled sets [15].

Domain shift and cross-system generalization represent another fundamental challenge limiting practical deployment. Variations in camera hardware, illumination, imaging geometry, material properties, and process parameters substantially alter visual defect appearance, causing trained models to exhibit degraded performance on different equipment/materials [7]. This is particularly problematic in industrial environments deploying multiple AM systems from different vendors, each with distinct imaging characteristics. Domain adaptation techniques, including adversarial training learning domain-invariant representations and few-shot learning enabling rapid adaptation with minimal additional labeled data, show promise [3]. Multi-source pre-training exposing models to diverse imaging conditions during initial training may improve generalization across heterogeneous environments.

Real-time processing requirements impose computational constraints carefully balanced against detection accuracy and model complexity. Real-time in-situ monitoring requires inference speeds compatible with high-frequency image acquisition during AM processes [2]. Modern YOLO-based models have demonstrated real-time inference on GPU platforms in controlled experimental settings [2]. However, deployment on resource-constrained embedded or edge platforms

remains challenging due to memory and computational limitations. Model optimization techniques such as quantization and pruning are commonly explored to reduce computational overhead while maintaining acceptable performance. Hybrid strategies combining fast screening with detailed analysis may offer promising speed-accuracy trade-offs.

Interpretability and explainability of deep learning-based models constitute increasingly important research directions, particularly for safety-critical applications in aerospace, medical device manufacturing, and regulated industries requiring decision transparency for validation and regulatory compliance [5]. Current deep learning models operate largely as black boxes, making it difficult to understand which image features drive classifications and whether models learn physically meaningful defect representations [5]. Attention mechanism visualization, gradient-based saliency mapping, and model-agnostic explanation techniques (LIME, SHAP) offer interpretation methods, though these remain limited in providing complete understanding of internal reasoning [12]. Physics-informed neural networks incorporating domain knowledge about AM processes and defect formation mechanisms into architectures represent promising directions for improving both interpretability and generalization, as these constrain learned representations to be consistent with known physical principles [11].

6.3 Future Research Priorities

Advancing AI-based defect detection requires coordinated research efforts across multiple temporal horizons and technical domains. Short-term priorities (1–2 years) should focus on establishing standardized benchmark datasets and evaluation protocols enabling systematic method comparison across diverse AM processes, materials, and defect types [1]. Developing publicly accessible datasets with comprehensive annotations covering multiple imaging modalities and process conditions would significantly accelerate progress by providing common evaluation grounds and reducing duplicated data collection [3]. Lightweight model architectures optimized for edge deployment represent another immediate priority, as industrial implementation increasingly demands real-time processing on cost-effective hardware integrated directly into AM systems without expensive server infrastructure [2]. Domain adaptation methodologies enabling rapid model fine-tuning with minimal additional labeled data should be prioritized to address generalization challenges limiting transferability across manufacturing systems [7].

Medium-term directions (2–5 years) should emphasize integrating AI-based detection with closed-loop process control architectures enabling automated corrective actions based on real-time quality assessments [2]. Such integration requires not only accurate defect detection but robust decision-making frameworks determining appropriate process parameter adjustments to mitigate defects without introducing new quality issues. Self-supervised learning approaches leveraging large volumes of unlabeled image data generated during normal manufacturing operations represent promising directions for reducing annotation requirements while improving robustness through exposure to diverse process

conditions. Multi-modal sensor fusion strategies combining visual imaging with thermal, acoustic, or complementary sensing modalities have demonstrated potential for improving detection accuracy and reducing false positives by capturing different physical signatures of defect formation processes [9]. Establishing industry-wide standards and best practices for AI-based quality assurance in AM, potentially through collaboration with standards organizations and regulatory bodies, would facilitate broader industrial adoption and enable systematic validation across application domains [7].

Long-term priorities (>5 years) should target transformative capabilities fundamentally expanding intelligent quality assurance scope and impact. Physics-informed neural network architectures integrating fundamental understanding of heat transfer, fluid dynamics, and solidification phenomena into model structures represent pathways toward more interpretable and physically consistent detection systems that can extrapolate beyond training data distributions [11]. Fully autonomous quality assurance frameworks integrating defect detection with predictive process modeling, adaptive control, and automated post-process inspection could enable lights-out manufacturing scenarios where AM systems continuously optimize their own operation based on real-time quality feedback without human intervention.

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